Ogbonna Arinze

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SUMMARY

Frontend Software Engineer with 8 years of experience building high-impact, user-centric web applications across healthcare, B2B SaaS, fintech, and civic tech environments. Deep expertise in **React** and **TypeScript**, with a product engineering mindset honed in startup and enterprise settings. Proven track record of owning frontend architecture end-to-end, from designing reusable component systems and design systems to optimizing performance and shipping features rapidly. Experienced with **AI-assisted development workflows**, production RAG/LLM systems, and fullstack contributions including **PostgreSQL**, NodeJS APIs, and **Docker**-based deployments.

CORE COMPETENCIES

- **Frontend Engineering:** React, TypeScript, Next.js, Vue.js, Redux, Zustand, Tailwind CSS, Radix UI, and scalable component/design systems.
- **Programming Languages:** TypeScript, modern JavaScript (ESNext), NodeJS, and Python.
- **Backend & Data:** NodeJS REST APIs, NestJS, PostgreSQL, MongoDB, Prisma, GraphQL, JWT authentication, and RBAC.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), embeddings, and agentic workflows.
- **Testing & Quality:** Playwright, Jest, Cypress, Storybook, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, GitHub Actions, CI/CD pipelines, Nx monorepo orchestration, and Vercel.
- **Performance & Observability:** Core Web Vitals, code splitting, CDN strategies, Lighthouse, bundle analysis, Sentry, and Datadog RUM.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, Figma design-to-code workflows, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, taking full ownership of the frontend architecture, optimizing workflows for clinical site integrations, and enhancing adoption.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query their data in real time via a conversational interface.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright** end-to-end tests into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface for a **B2B SaaS** platform, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and more confident feature shipping.

- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices with precise user story planning and estimates.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS**, and **Axios**, owning features end-to-end from conception through deployment.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas in a fast-paced startup environment.
- Transformed **Figma** designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to Heroku and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing REST API endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with **React 19** and **Tailwind CSS v4** (22 interactive chart types via Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built a **multi-agent AI system** using the Mastra framework with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and Cohere reranking.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.
- Implemented **SSE streaming** for real-time chat responses, **PostgreSQL 16** with **Prisma v7**, phone OTP authentication, Telegram bot with webhook handling, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Integrated **Langfuse** for LLM observability, **Infisical** for secrets management across environments, and **Traefik** for reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*